

Bayesian inference on the Planck 2018 binned temperature (TT) power spectrum

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Abstract

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1 Introduction and background

Introduction placeholder.

2 Data

We use the publicly released Planck 2018 binned temperature power spectrum, distributed as the plain-text file `COM_PowerSpect_CMB-TT-binned_R3.01.txt` in the Planck Public Data Release 3 ancillary data [1]. The file contains the CMB-only binned bandpowers, marginalised over astrophysical foregrounds and released for plotting and inspection purposes. It has five whitespace separated columns:

Column	Meaning
ℓ	effective (weighted) multipole of the bin
D_ℓ	bandpower $D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell$ [μK^2]
$-dD_\ell$	lower 1σ uncertainty on D_ℓ
$+dD_\ell$	upper 1σ uncertainty on D_ℓ
BestFit	Planck best-fit ΛCDM bandpower

The spectrum spans roughly $\ell \simeq 30$ to $\ell \simeq 2500$ in a few tens of bins. Throughout we work directly with D_ℓ in μK^2 and treat each bin's uncertainty as independent (diagonal covariance); the consequences of this simplification are discussed in Section 4. Figure 1 shows the loaded data.

Figure 1: Planck 2018 binned TT bandpowers D_ℓ with 1σ error bars (placeholder – insert `figures/planck_tt_data.pdf` produced by the notebook).

3 Theoretical model

The theoretical TT power spectrum is computed with the Boltzmann code CAMB [3] through its Python interface (the `camb` package, installed via `pip`; no manual compilation is required). The model is parametrised by the six base ΛCDM parameters

$$\theta = (\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s), \quad (1)$$

where $\Omega_b h^2$ and $\Omega_c h^2$ are the physical baryon and cold dark matter densities, H_0 the Hubble constant, τ the reionisation optical depth, A_s the amplitude and n_s the tilt of the primordial scalar power spectrum. We sample $\ln(10^{10} A_s)$ rather than A_s because it varies over a more comparable numerical range.

For a given θ , CAMB returns the lensed spectrum $D_\ell^{TT} = \ell(\ell+1)C_\ell^{TT}/2\pi$ in μK^2 at integer multipoles. To compare with the binned data we evaluate the theory at the *effective* multipoles ℓ_b of each bin by linear interpolation,

$$D_b^{\text{th}}(\theta) = D_\ell^{TT}(\theta)\Big|_{\ell=\ell_b}, \quad (2)$$

which approximates the Planck bin window functions (not provided in the public binned file) and is adequate because D_ℓ^{TT} varies smoothly across each bin.

Two deliberate simplifications are adopted for computational speed, both negligible at the precision of this simplified analysis: neutrinos are treated as massless ($\sum m_\nu = 0$ instead of the default 0.06 eV), and the lensing potential is computed at low accuracy. Figure 2 overlays the CAMB model evaluated at the Planck 2018 fiducial parameters on the data as a sanity check.

Figure 2: CAMB model at the Planck 2018 fiducial parameters overlaid on the binned TT data (placeholder – insert `figures/theory_vs_data.pdf` from the notebook).

4 Methodology

Methodology placeholder.

5 Results

Results placeholder.

6 Conclusions

Conclusions placeholder.

7 Declaration of AI use

This project used the AI assistant *Claude (Anthropic)* as a coding and writing aid. Concretely, it was used to: (i) propose the project structure, (ii) draft and clean the Python code that loads the data, builds the theoretical model, defines the likelihood and runs the sampler, and (iii) help phrase parts of this report. Every code cell was read, executed and verified by the author, and all scientific choices (model, priors, simplifications) are the author’s responsibility. The exact (hypothetical) prompts that would reproduce each code block are logged verbatim in Appendix A, so the degree and nature of AI assistance is fully transparent.

A Hypothetical prompts log

This appendix records, step by step, a prompt that would generate the corresponding code. It doubles as the transparency record for Section 7.

Step 1 — Data loading and inspection

Hypothetical prompt

“In Python, write a short, clean script (NumPy + Matplotlib only) that loads the Planck 2018 binned TT power spectrum from the local file `data/COM_PowerSpect_CMB-TT-binned_R3.01.txt`, downloading it once from the IRSA Planck PR3 mirror if it is not present. The file has five whitespace-separated columns: effective multipole ℓ , bandpower $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ in μK^2 , lower and upper 1σ errors, and the Planck best-fit. Return ℓ , D_ℓ and a symmetric error array, then plot D_ℓ versus ℓ with error bars, labelled axes and a saved `.pdf` figure. Keep comments short.”

Step 2 — Theoretical model with CAMB

Hypothetical prompt

“Using the `camb` Python package, write a function `theory_dl(theta)` that returns the lensed TT power spectrum $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ in μK^2 , evaluated at the effective multipoles of the Planck binned data by linear interpolation. `theta` is the six-vector $[\Omega_b h^2, \Omega_c h^2, H_0, \tau, \ln(10^{10} A_s), n_s]$; convert $\ln(10^{10} A_s)$ back to A_s . For speed set massless neutrinos and low lensing-potential accuracy, and compute CAMB only slightly beyond the maximum data multipole. Then plot the model at the Planck 2018 fiducial parameters on top of the data and save a `.pdf`. Keep the code short with brief comments.”

References

- [1] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, A&A 641, A6 (2020).
- [2] D. Foreman-Mackey et al., *emcee: The MCMC Hammer*, PASP 125, 306 (2013).
- [3] A. Lewis, A. Challinor, A. Lasenby, *Efficient Computation of CMB anisotropies in closed FRW models*, ApJ 538, 473 (2000).